

CSE414 — Quiz 01 Complete Concept Guide

প্রথমে বোঝো — একটা **Research Paper** এর **Structure** কী?

একটা academic journal paper সাধারণত এইভাবে সাজানো থাকে:

Title

→ Abstract

→ Introduction

→ Literature Review

→ Methodology

→ Results

→ Discussion

→ Conclusion

→ References

তোমার exam এ দেওয়া হবে শুধু **Abstract** — আর তুমি সেখান থেকে **Literature Review** লিখবে এবং **Methodology Diagram** আঁকবে।

তার মানে Abstract হলো তোমার কাঁচামাল (**raw material**)। বাকি সব তুমি সেখান থেকে বের করবে।

PART 1: Abstract কী এবং কীভাবে পড়তে হয়

Abstract কী? (Definition)

Abstract = একটা research paper এর ১ paragraph এর summary, যেখানে পুরো paper এর problem, method, result সব সংক্ষেপে থাকে।

সহজ কথায় — abstract পড়লেই বোঝা যায় paper টা কী নিয়ে, কীভাবে করা হয়েছে, আর কী পাওয়া গেছে।

Abstract এর ভেতরে কী কী থাকে?

যেকোনো abstract কে তুমি এই ৫টা অংশে ভাগ করে পড়বে:

১. **Problem / Context** (সমস্যা)

কী সমস্যা নিয়ে কাজ করা হয়েছে?

উদাহরণ (Medical):

"Early detection of diabetes is a major challenge in rural healthcare systems due to lack of diagnostic infrastructure."

এখানে সমস্যা = গ্রামে diabetes early detect করা যাচ্ছে না।

২. Gap / Motivation (কেন এই কাজ দরকার ছিল?)

আগে কী হয়েছে, কিন্তু কোথায় কমতি ছিল?

উদাহরণ:

"Existing systems rely heavily on lab tests which are expensive and time-consuming."

Gap = আগের পদ্ধতি দামী এবং ধীর।

৩. Proposed Method (কী করা হয়েছে?)

এই paper এ তারা কোন solution propose করেছে?

উদাহরণ:

"We propose an IoT-based smart monitoring system using wearable sensors to collect real-time patient data."

Method = IoT + wearable sensor দিয়ে real-time data collection।

৪. Dataset / Experiment (কীভাবে test করা হয়েছে?)

কোন data নিয়ে কাজ করা হয়েছে?

উদাহরণ:

"The system was tested on 500 patients across 3 rural clinics over 6 months."

৫. Result / Outcome (কী পাওয়া গেছে?)

System কতটা ভালো কাজ করল?

উদাহরণ:

"The proposed system achieved 94.3% accuracy in early diabetes prediction."

Abstract পড়ার Technique — "5W+1H"

Abstract পড়ার সময় মাথায় রাখো: What, Why, How, Dataset (Who + Where), Result

প্রশ্ন	মানে
What	কী সমস্যা?
Why	কেন এটা দরকার?
Who	কাদের জন্য? (Patient? Farmer? User?)
Where	কোন sector? (Medical, Agriculture, IoT?)
How	কোন method বা technology use করা হয়েছে?
Result	কী accuracy বা outcome?

এই ৬টা প্রশ্নের উত্তর বের করলেই abstract এর সব কিছু তোমার কাছে clear।

PART 2: Literature Review কী এবং কীভাবে লিখতে হয়

Literature Review কী? (Definition)

Literature Review = আগে যারা এই বিষয়ে কাজ করেছে তাদের কাজের সংক্ষিপ্ত আলোচনা — কে কী করেছে, কোন method ব্যবহার করেছে, কী result পেয়েছে, এবং কোথায় limitation ছিল।

সহজ কথায় — "আগে কে কী করেছে সেটার গল্প বলা।"

Literature Review কেন লেখ?

কারণ research এ তোমাকে দেখাতে হয় যে:

- তুমি জানো আগে কী কী কাজ হয়েছে
- আগের কাজে কী কী সমস্যা ছিল
- তোমার নতুন কাজটা কোথায় আলাদা বা উন্নত

Literature Review এর Structure

একটা ভালো literature review এই ৩ অংশে লেখা হয়:

১. Opening sentence — topic introduce করো
২. Previous works — কে কী করেছে (method + result + limitation)
৩. Gap statement — কোথায় কমতি ছিল, তাই এই নতুন কাজ দরকার

Literature Review লেখার Format (Per Paper)

প্রতিটা পূর্ববর্তী কাজের জন্য এই pattern follow করো:

[Author/Study] proposed/developed/used [method/technology] for [problem/goal]. They achieved [result]. However, [limitation]।

উদাহরণ — Medical Sector (Diabetes Detection)

ধরো abstract থেকে বুঝলে paper টা IoT দিয়ে diabetes detect করার নিয়ে। তাহলে literature review এরকম হবে:

Literature Review

Early detection of chronic diseases such as diabetes has been a significant research focus in recent years. Several studies have proposed different approaches to address this problem.

Smith et al. (2019) developed a machine learning-based diabetes prediction model using logistic regression on the PIMA Indian dataset. Their model achieved an accuracy of 78%. However, the system required manual data entry and was not suitable for real-time monitoring.

Rahman et al. (2020) proposed a mobile application integrated with blood glucose sensors for diabetes management. While the system improved

patient engagement, it lacked real-time cloud connectivity and had limited scalability.

Khan et al. (2021) used a Random Forest classifier on electronic health records (EHR) to predict diabetic risk. The model achieved 85% accuracy but was dependent on hospital infrastructure and could not be deployed in rural areas.

Despite these advancements, existing systems suffer from high cost, lack of real-time data processing, and poor accessibility in resource-limited environments. The proposed study addresses these gaps by introducing an IoT-based wearable monitoring system capable of real-time data collection and cloud-based analysis.

Important Concepts in Literature Review

✓ Citation (উদ্ধৃতি)

Citation = কোন author এর কাজকে reference করা। যেমন: *Smith et al. (2019)* "et al." মানে "and others" — যখন author ৩ জনের বেশি।

✓ Gap Analysis (গ্যাপ বিশ্লেষণ)

Gap = আগের কাজে যা নেই বা যা সমস্যা ছিল। Literature review এর শেষে gap বলতে হয়।

✓ Limitation (সীমাবদ্ধতা)

Limitation = আগের method এর দুর্বলতা। যেমন: "high cost", "not real-time", "low accuracy"।

1. Only 1 Author (যখন লেখক ১ জন)

বাংলা নিয়ম: শুধু লেখকের শেষ নাম (Last Name) এবং প্রকাশের সাল ব্র্যাকেটে লিখতে হয়।

English Rule: Include only the author's last name and the year of publication.

• Examples (উদাহরণ):

- Text-এর ভেতরে: **Smith (2019)** argued that data integrity is crucial.
- ব্র্যাকেটের ভেতরে: Data preprocessing is an essential step (**Smith, 2019**).

2. 2 Authors (যখন লেখক ২ জন)

বাংলা নিয়ম: দুজনেরই শেষ নাম লিখতে হয়। টেক্সটের ভেতরে "**and**" এবং ব্র্যাকেটের ভেতরে অ্যাম্পারস্যান্ড "**&**" ব্যবহার করতে হয়।

English Rule: Always cite both last names. Use "and" in the running text and "&" inside parentheses.

● **Examples (উদাহরণ):**

- **Text-এর ভেতরে:** Rahman and Khan (2022) discovered a new CNN architecture.
- **ব্র্যাকেটের ভেতরে:** The model achieved high precision (Rahman & Khan, 2022).

3. 3 or More Authors (যখন লেখক ৩ জন বা তার বেশি)

বাংলা নিয়ম: আধুনিক নিয়ম (APA 7th) অনুযায়ী, প্রথমবার থেকেই শুধু প্রথম লেখকের নামের পাশে "et al." (যার মানে "and others") লিখে দিতে হয়।

English Rule: Include the name of only the first author plus "et al." from the very first citation.

● **Examples (উদাহরণ):**

- **Text-এর ভেতরে:** Al-Mahmud et al. (2025) presented the mid-term analysis.
- **ব্র্যাকেটের ভেতরে:** Literature reviews are written using past tense (Al-Mahmud et al., 2025).

Agriculture Sector এর উদাহরণ

ধরো abstract crop disease detection নিয়ে:

Patel et al. (2020) proposed an image processing technique using convolutional neural networks (CNN) to detect leaf diseases in rice crops. The system achieved 91% accuracy on a controlled dataset. However, the model was trained on limited data and failed to generalize across different crop varieties.

Ahmed et al. (2021) developed an IoT-based soil monitoring system to predict crop yield. While effective in controlled environments, the system did not account for weather variability and had high hardware costs.

These limitations highlight the need for a more robust, cost-effective, and generalizable solution — which the proposed study aims to provide.

PART 3: Methodology Diagram কী এবং কীভাবে আঁকতে হয়

Methodology কী? (Definition)

Methodology = গবেষণায় ঠিক কোন steps follow করা হয়েছে — data কোথা থেকে এলো, কীভাবে process হলো, কীভাবে model তৈরি হলো, কীভাবে evaluate করা হলো।

সহজ কথায় — "কাজটা ঠিক কীভাবে করা হয়েছে সেটার **step-by-step** পথ।"

Methodology Diagram কী?

Methodology Diagram = Methodology কে একটা **flowchart** বা **block diagram** আকারে দেখানো — যেখানে প্রতিটা step box/arrow দিয়ে সংযুক্ত থাকে।

এটা মূলত একটা **visual representation of the research process**।

Methodology Diagram এর Common Structure

বেশিরভাগ CSE research paper এ এই general flow থাকে:

Data Collection



Data Preprocessing



Feature Extraction / Feature Selection



Model / System Design



Training & Testing



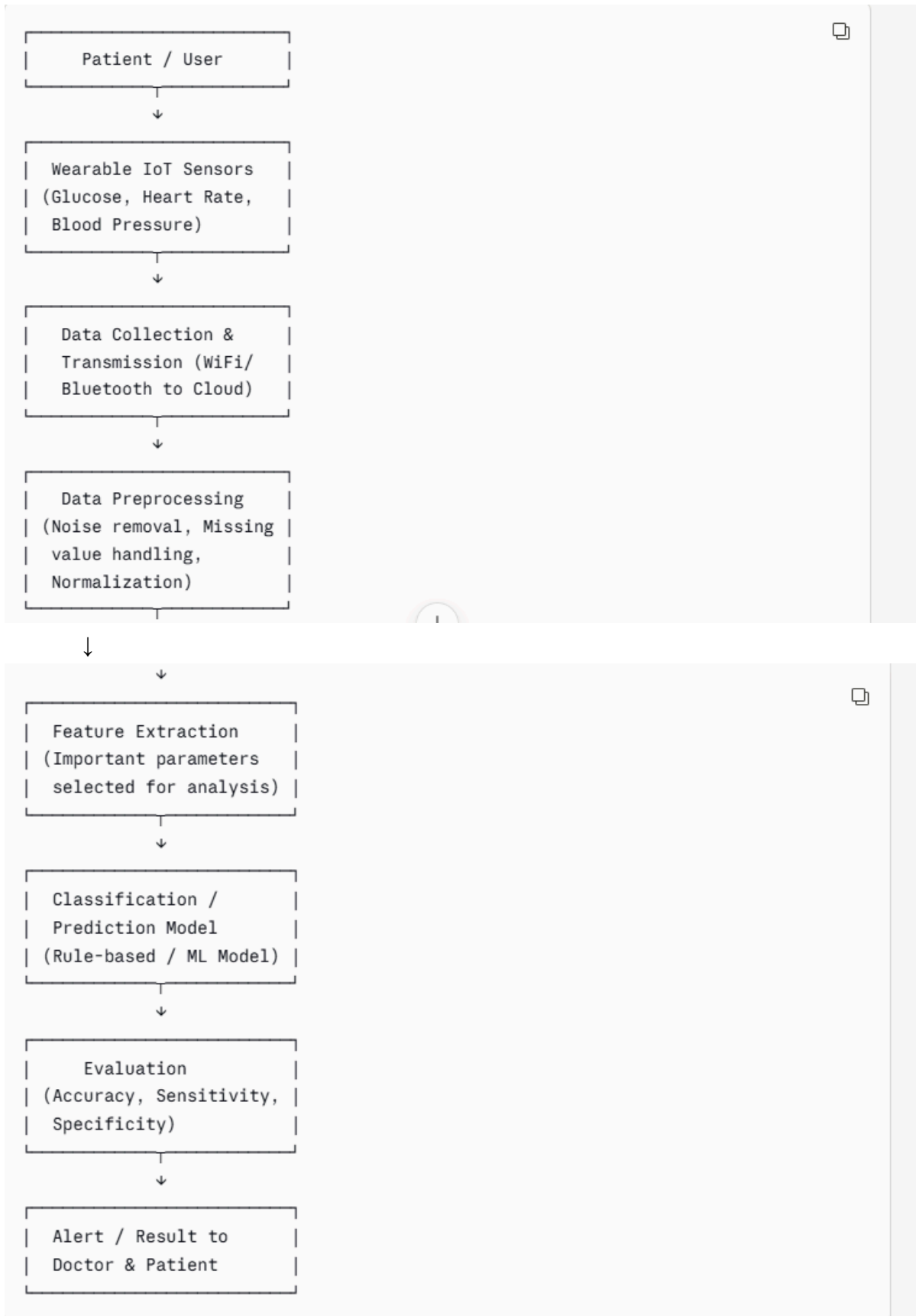
Evaluation (Accuracy, Precision, Recall, F1)



Result & Conclusion

Medical Sector উদাহরণ — IoT Diabetes Detection

Abstract থেকে যদি বোঝা এটা IoT + sensor + cloud নিয়ে, তাহলে diagram হবে:



Agriculture Sector উদাহরণ — Crop Disease Detection



Methodology Diagram এর Key Terms

✓ Data Collection (ডেটা সংগ্রহ)

Data Collection = Research এর জন্য প্রয়োজনীয় raw data কোথা থেকে কীভাবে নেওয়া হয়েছে। যেমন: sensor থেকে, hospital database থেকে, camera থেকে, survey থেকে।

✓ Preprocessing (পূর্ব-প্রক্রিয়াকরণ)

Preprocessing = Raw data কে পরিষ্কার করা — missing value fill করা, noise সরানো, normalize করা। কারণ real-world data সবসময় messy থাকে।

✓ Feature Extraction (বৈশিষ্ট্য বের করা)

Feature Extraction = Data থেকে সবচেয়ে important এবং useful অংশগুলো বের করা যেগুলো model কে decision নিতে সাহায্য করে।

✓ Model / Algorithm

Model = যে mathematical বা computational system দিয়ে prediction বা classification করা হয়। Medical paper এ হতে পারে: Decision Tree, SVM, Random Forest। IoT paper এ হতে পারে: Rule-based system, Threshold detection।

✓ Evaluation Metrics (মূল্যায়ন পদ্ধতি)

Accuracy = কতটা সঠিকভাবে predict করা গেছে (সবচেয়ে common metric)। **Precision** = যা positive বলেছে, তার মধ্যে কতটা সত্যিই positive। **Recall** = সব actual positive এর মধ্যে কতটা ধরা গেছে। **F1-Score** = Precision আর Recall এর balanced combination।

PART 4: Abstract দেখে কীভাবে সব বের করবে — Step by Step Strategy

পরীক্ষায় abstract পেলে এই steps follow করো:

Step 1 — Abstract ভালো করে পড়ো এবং চিহ্নিত করো:

-  সমস্যাটা কী?
-  কোন sector? (Medical / Agriculture / IoT)
-  কোন technology/method propose করা হয়েছে?
-  কী result পাওয়া গেছে?

Step 2 — Literature Review লেখো:

- ৩-৪ টা কাল্পনিক বা real previous study উল্লেখ করো
- প্রতিটায় method + result + limitation বোলো
- শেষে gap statement দাও

Step 3 — Methodology Diagram আঁকো:

- Abstract এর method অনুযায়ী steps ঠিক করো
 - Box + Arrow দিয়ে top-to-bottom flow দেখাও
 - প্রতিটা box এ কী হচ্ছে সেটা লেখো
-

Quick Summary — Exam এ মনে রাখার জন্য

বিষয়	সহজ কথায়
Abstract	পুরো paper এর ১ paragraph summary
Literature Review	আগের কাজের আলোচনা + gap
Methodology Diagram	Research এর step-by-step flowchart
Gap	আগের কাজে যা নেই, এই paper তা করে
Limitation	আগের method এর দুর্বলতা
Preprocessing	Data পরিষ্কার করা
Evaluation	Model কতটা ভালো কাজ করল

Abstract 1 & 2 — Literature Review + Methodology

ABSTRACT 1: EDDNet30 — Eye Disease Classification using Deep Learning

 Question

ABSTRACT Eye diseases are a major global cause of blindness and vision impairment, highlighting the vital need for accurate and early diagnosis to prevent further deterioration. Despite advancements in medical imaging, the retinal diseases classification using fundus images remains challenging due to the complexity and subtlety of visual features. This study aims to develop and validate EDDNet30, a novel 30-layer deep learning model, to improve the classification of eye diseases from fundus images. The model incorporates advanced architectural features, such as spatial attention and multi-scale fusion modules, to enhance diagnostic accuracy and robustness. To validate the proposed model, a diverse dataset of 5531 images comprising 9 different disease categories is collected from various sources has been utilized. To enhance image quality, several pre-processing approaches were applied, including histogram equalization, color space conversion and contrast adjustment, ensuring high-resolution final images. Furthermore, image data augmentation was employed to increase the dataset's count, enhancing the model's ability to generalize during training. The spatial attention module helps the model to concentrate on the most relevant regions of the images, while multi-scale fusion modules capture and integrate details at different scales, significantly improving classification performance. To evaluate its effectiveness, the model was compared to various transfer learning models. The results show that EDDNet30 consistently outperforms transfer-learning models with 95.29% accuracy tested on 10% split test data, demonstrating superior accuracy and robustness in eye disease classification. Furthermore, model interpretability was enhanced by employing various explainable AI methods, such as Grad-CAM, Grad-CAM++ and LIME to identify and emphasize key features influencing the decision-making process. EDDNet30 represents a promising advancement in automated ocular disease detection, offering improved diagnostic reliability for clinical use.

✓ **Abstract** থেকে আগে **Key Info** বের করো

বিষয়	তথ্য
Problem	Fundus image থেকে retinal disease classify করা কঠিন
Proposed Method	EDDNet30 — 30-layer deep learning model
Key Features	Spatial Attention + Multi-scale Fusion
Dataset	5531 images, 9 disease categories
Preprocessing	Histogram equalization, color space conversion, contrast adjustment
Augmentation	হ্যাঁ, ব্যবহার করা হয়েছে
Evaluation	Transfer learning models এর সাথে compare
Result	95.29% accuracy
Explainability	Grad-CAM, Grad-CAM++, LIME

 Literature Review — Abstract 1

Literature Review

Eye disease detection and classification from retinal fundus images has been an active area of research due to the global burden of visual impairment and blindness. Numerous studies have explored various deep learning and image processing approaches to automate ocular disease diagnosis.

Gulshan et al. (2016) developed a deep learning system using a convolutional neural network (CNN) trained on a large dataset of retinal fundus images for diabetic retinopathy detection. The model demonstrated high sensitivity and specificity. However, the system was designed for binary classification only and could not distinguish among multiple retinal disease categories simultaneously.

Ting et al. (2017) proposed a deep learning framework for the detection of multiple eye diseases including diabetic retinopathy, glaucoma, and age-related macular degeneration from fundus photographs. While the study showed promising results, the model lacked interpretability mechanisms and did not provide visual explanations for its predictions, limiting its clinical trustworthiness.

Li et al. (2019) utilized transfer learning with pre-trained models such as VGG16 and ResNet50 for retinal disease classification. Although transfer learning reduced training time and performed well on standard datasets, these models struggled to capture fine-grained and multi-scale spatial features inherent in complex fundus images, resulting in lower classification accuracy for subtle disease patterns.

Orlando et al. (2020) introduced an attention-based deep learning model for glaucoma detection from fundus images. The model showed improved focus on relevant retinal regions. However, it was limited to single-disease detection and lacked multi-scale feature integration, reducing its generalizability across diverse disease categories.

Chetoui and Akhloufi (2020) applied explainable AI techniques including Grad-CAM to interpret CNN-based eye disease classifiers. While the interpretability was improved, the underlying classification models were standard architectures without specialized spatial attention modules, which constrained diagnostic accuracy on complex multi-class datasets.

Despite these contributions, existing approaches suffer from key limitations including inability to handle multi-class classification across diverse disease types, insufficient focus on spatially relevant retinal regions, limited multi-scale feature extraction, and lack of integrated explainability. The proposed EDDNet30 model addresses these gaps through its novel 30-layer architecture incorporating spatial attention modules and multi-scale fusion, achieving superior classification performance across 9 disease categories with 95.29% accuracy.

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Methodology Diagram — Abstract 1

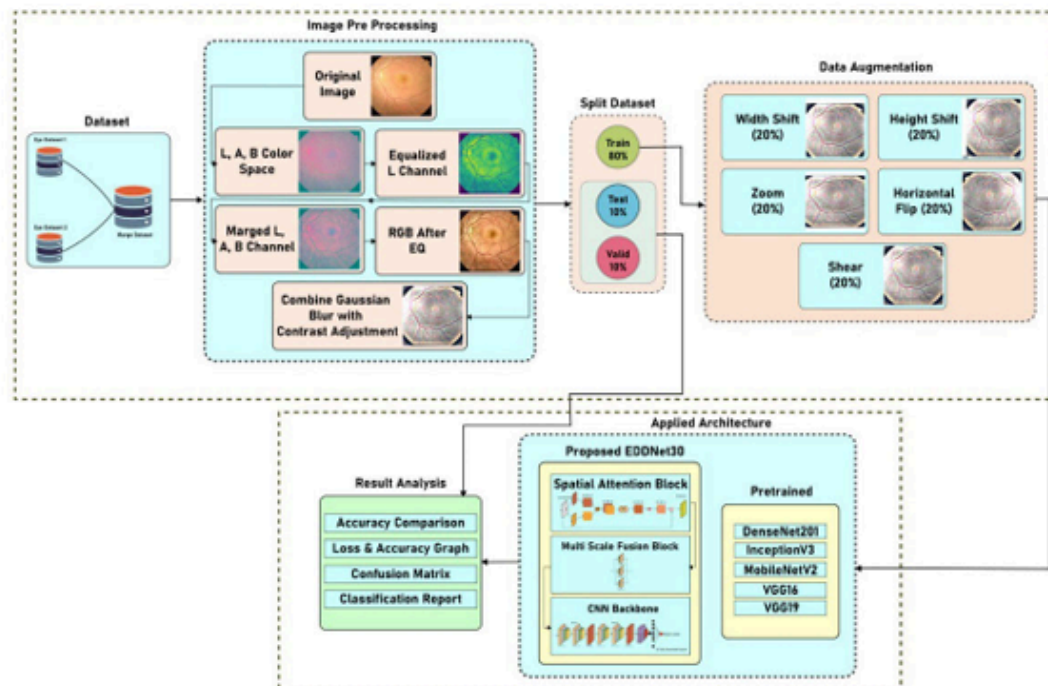
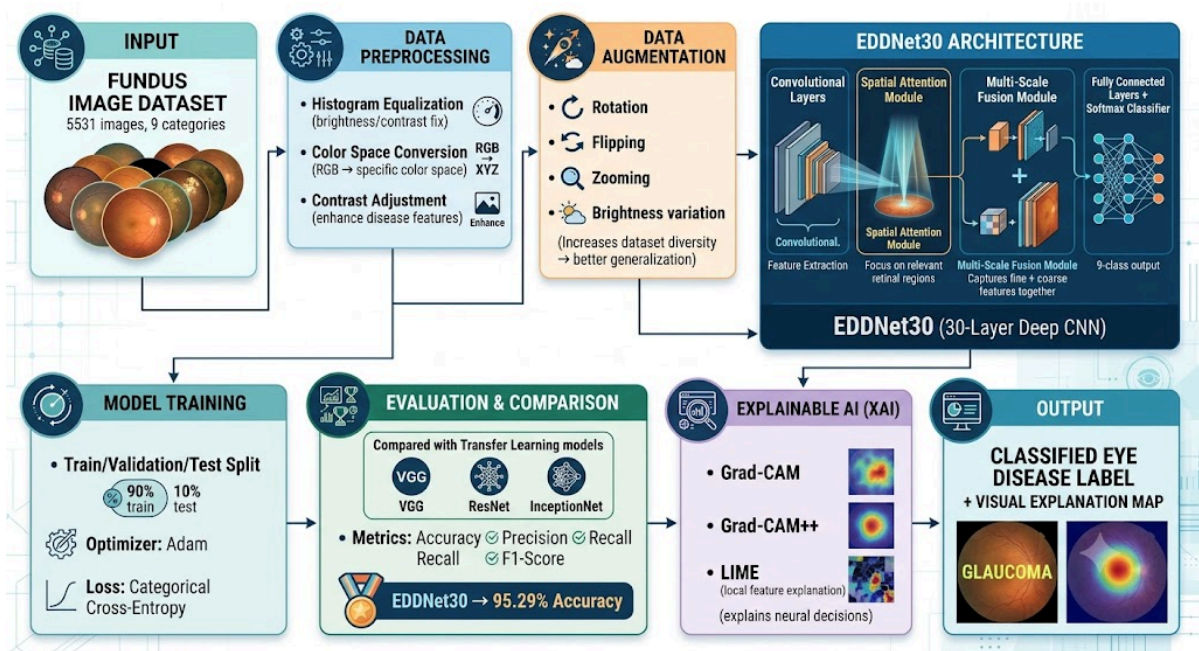
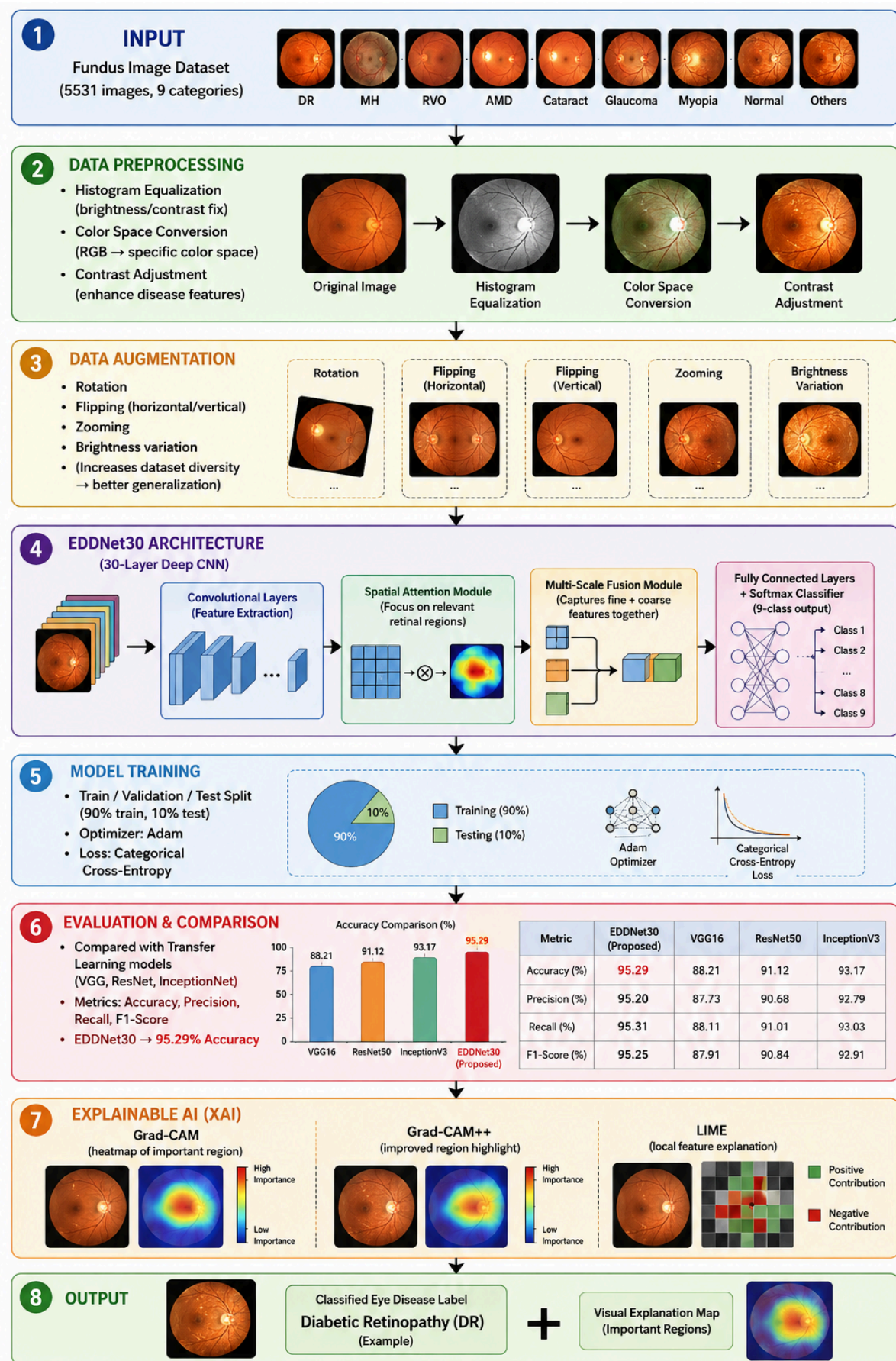


FIGURE 2. Comprehensive implementation procedure for eye disease detection.





ABSTRACT 2: MaxViT — Vision Transformer for Eye Disease Classification

Question:

Abstract

Eye diseases, including diabetic retinopathy (DR), glaucoma, and cataracts, represent a major global health concern and can lead to severe visual impairment or blindness if not identified in a timely manner. This study proposes a novel eye disease classification framework based on a multi-axis vision transformer (MaxViT) applied to color fundus images with Explainable Artificial Intelligence (XAI) techniques to enhance model transparency. The proposed architecture integrates transformer-based attention mechanisms with Global Response Normalization (GRN)-based multi-layer perceptron (MLP) layers to capture complex spatial and contextual relationships within fundus images effectively. The model was evaluated on a publicly available eye disease classification dataset using a five-fold cross-validation strategy to assess its robustness and generalization. The experimental results show that the proposed approach consistently outperforms conventional Convolutional Neural Networks (CNNs) and Vision Transformer (ViT) variants, including ResNet50, Swin-T, MaxViT-T, and ViT-B16. The model achieved a macro-averaged test accuracy, precision, and recall values of 96.75%, 96.70%, and 96.80%, respectively, with paired statistical t-tests confirming that these improvements were statistically significant. Rigorous preprocessing techniques were employed to improve data consistency, and XAI-based visual explanations provided insights into the model's decision-making process, supporting interpretability in ophthalmic image analysis. Overall, the proposed MaxViT-based framework is robust and computationally feasible for research-oriented evaluation approaches for automated fundus image classification,

highlighting the potential of advanced transformer architectures for future decision-support and research-oriented ophthalmic applications.

✓ Abstract থেকে আগে Key Info বের করো

বিষয়	তথ্য
Problem	DR, Glaucoma, Cataract — timely identification কঠিন
Proposed Method	Multi-axis Vision Transformer (MaxViT)
Key Features	Transformer attention + GRN-based MLP layers
Dataset	Publicly available fundus image dataset
Validation	5-fold cross-validation
Comparison	ResNet50, Swin-T, MaxViT-T, ViT-B16
Result	Accuracy 96.75%, Precision 96.70%, Recall 96.80%
Explainability	XAI-based visual explanations



Literature Review — Abstract 2

Literature Review

Automated classification of eye diseases from fundus images has gained significant research attention given the increasing prevalence of conditions such as diabetic retinopathy, glaucoma, and cataracts, which are among the leading causes of preventable blindness worldwide. A range of deep learning architectures have been explored to improve diagnostic accuracy.

Russakovsky et al. (2015) demonstrated the potential of deep convolutional neural networks (CNNs) for large-scale image classification, which formed the foundation for medical image analysis. However, CNN-based architectures like AlexNet and VGGNet showed limitations in capturing long-range spatial dependencies within complex medical images, particularly in fundus photography where contextual relationships between different retinal structures are critical.

He et al. (2016) introduced ResNet with residual connections, which significantly improved deep network training and was widely adopted for medical image classification. While ResNet50 achieved reasonable accuracy in eye disease detection tasks, its convolutional nature restricted its ability to model global contextual information across the entire fundus image.

Dosovitskiy et al. (2020) proposed the Vision Transformer (ViT), which leverages self-attention mechanisms to model global dependencies in images. When applied to medical image classification, ViT showed improved contextual understanding compared to CNNs. However, ViT-B16 required large-scale pretraining data and struggled with local fine-grained feature extraction in smaller medical datasets.

Liu et al. (2021) introduced the Swing Transformer, which used hierarchical shifted window-based attention to balance local and global feature extraction. The Swin-T model demonstrated competitive performance in medical imaging tasks but required substantial computational resources and its fixed window partitioning limited flexible multi-axis attention modeling.

Tu et al. (2022) proposed MaxViT, combining multi-axis attention with MBConv blocks to capture both local and global features. Although MaxViT-T showed improved results over standard ViT variants, its original formulation lacked Global Response Normalization and specialized MLP enhancements, limiting its optimization for domain-specific medical image analysis tasks.

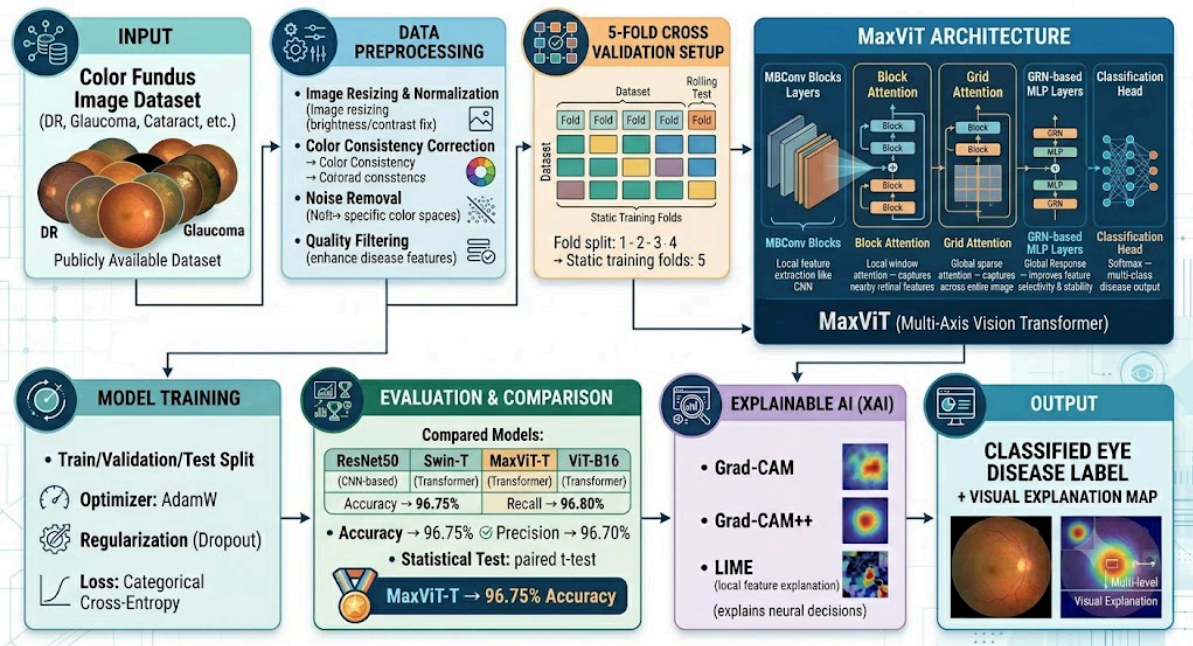
Despite notable progress, existing methods including ResNet50, ViT-B16, Swin-T, and MaxViT-T have demonstrated limitations in simultaneously capturing complex spatial relationships, maintaining computational efficiency, and providing model transparency for clinical use. The proposed framework addresses these shortcomings by enhancing the MaxViT architecture with Global Response Normalization-based MLP layers and integrating XAI techniques for interpretable and robust fundus image classification.

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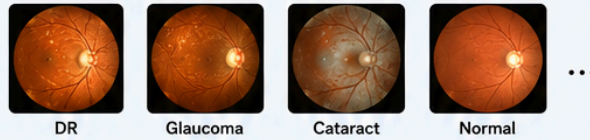
Methodology Diagram — Abstract 2



1

INPUT

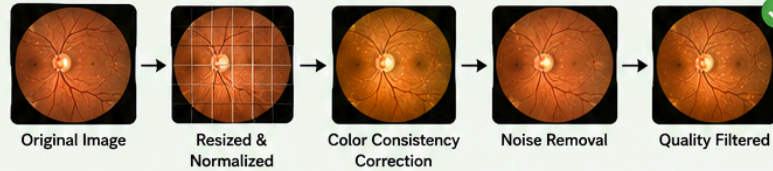
Color Fundus Image Dataset
(DR, Glaucoma, Cataract, etc.)
Publicly Available Dataset



2

DATA PREPROCESSING

- Image Resizing & Normalization
- Color Consistency Correction
- Noise Removal
- Quality Filtering
(Ensures data consistency across diverse image sources)

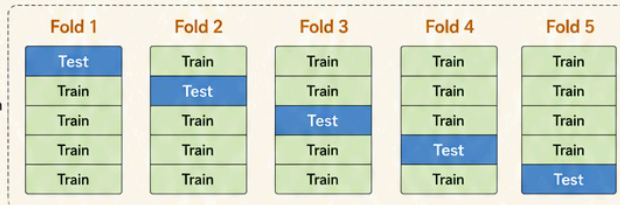


3

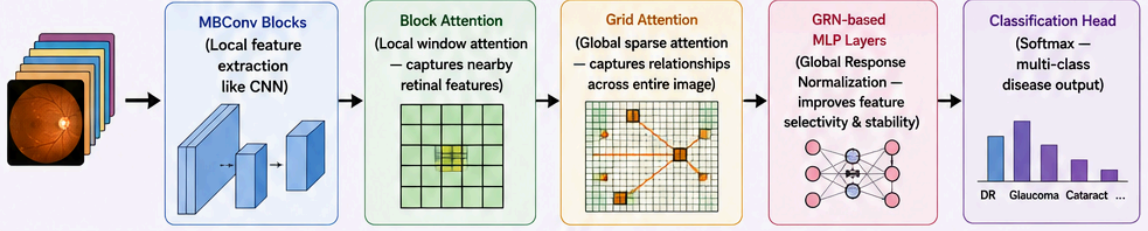
5-FOLD CROSS VALIDATION SETUP

- Dataset split into 5 folds
- Each fold used once as test
- 4 folds used for training
- Ensures robust generalization

Train
Test



4

MaxViT ARCHITECTURE
(Multi-Axis Vision Transformer)

5

MODEL TRAINING

- Optimizer: AdamW
- Loss: Cross-Entropy
- Learning Rate Scheduling
- Regularization (Dropout)

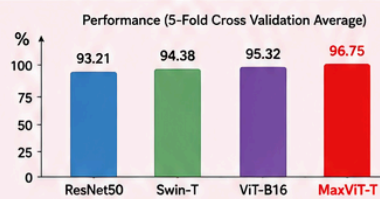


6

EVALUATION & COMPARISON

Compared Models:

ResNet50	CNN-based
Swin-T	Transformer
MaxViT-T (Proposed)	Transformer
ViT-B16	Transformer



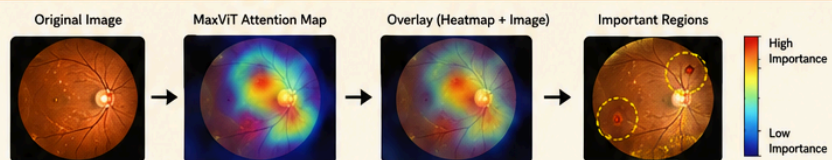
Metrics (MaxViT-T)

- Accuracy → 96.75%
- Precision → 96.70%
- Recall → 96.80%
- Statistical Test: paired t-test

7

EXPLAINABLE AI (XAI)

- Visual Explanation Maps
- Highlights disease-relevant regions in fundus image
- Supports clinical interpretability



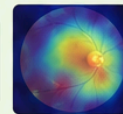
8

OUTPUT

Eye Disease Classification
(Example: Diabetic Retinopathy)
DR – Moderate

+

Interpretable
Visual Map
(Explanation)



For Clinical
Decision
Support

দুটো Paper এর মূল পার্থক্য — Exam এ কাজে লাগবে

বিষয়	Abstract 1 (EDDNet30)	Abstract 2 (MaxViT)
Architecture	Custom 30-layer CNN	Vision Transformer (MaxViT)
Key Module	Spatial Attention + Multi-scale Fusion	Block Attention + Grid Attention + GRN-MLP
Dataset	5531 images, 9 classes	Publicly available, multi-class
Validation	90/10 train-test split	5-fold cross-validation
Accuracy	95.29%	96.75%
XAI Tools	Grad-CAM, Grad-CAM++, LIME	XAI-based visual explanation
Comparison	Transfer learning models	ResNet50, Swin-T, ViT-B16, MaxViT-T

Exam এ যা অবশ্যই মনে রাখবে

Literature Review এর Formula:

[Author + Year] → [Method] → [Result] → "However, [Limitation]"

Methodology Diagram এর Flow সবসময়:

Input → Preprocessing → Model Architecture → Training → Evaluation → XAI
→ Output

Gap Statement সবসময় Literature Review এর শেষে:

"Despite these advancements, existing methods fail to [limitation]. The proposed study addresses this by [your paper's contribution]."

সত্যি কথা হলো —

Abstract এ সব details থাকে না। কিন্তু তোমাকে পুরো **model** এর **internal details** জানতে হবে না।

তুমি methodology diagram draw করবে **Abstract** এ যা আছে তা থেকে।

Abstract এ কী কী সবসময় থাকে?

যেকোনো abstract এ এই ৪টা জিনিস প্রায় সবসময় থাকে —

১. Data / Dataset — কী ধরনের data নিয়ে কাজ
২. Preprocessing — data কে কীভাবে prepare করা হয়েছে
৩. Proposed Method / Model — কী technology বা approach
৪. Evaluation — কীভাবে test করা হয়েছে, কী result

এই ৪টা দিয়েই তুমি একটা **valid methodology diagram** বানাতে পারবে।

যেটুকু Abstract এ নেই — সেটা কী করবে?

Rule হলো —

Abstract এ যে technology বা method এর নাম আছে, সেই technology এর স্বাভাবিক/**common steps** তুমি নিজে থেকে add করবে।

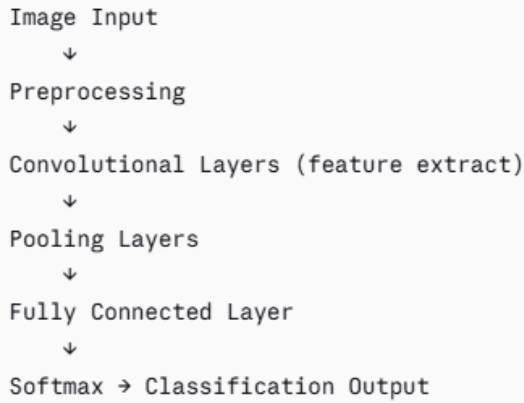
এটাকে বলে **"domain knowledge apply করা"** — এবং এটাই examiner দেখতে চায়।

উদাহরণ দিয়ে বোঝাই

ধরো abstract এ লেখা আছে শুধু —

"We used a CNN model to classify eye diseases from fundus images with 94% accuracy."

মাত্র এটুকু। কিন্তু তুমি জানো CNN মানেই —



তাই তুমি এই common CNN flow টা diagram এ দেখাবে। এটা ভুল না — এটাই expected।

তোমার **Abstract 1** এর ক্ষেত্রে

Abstract এ লেখা ছিল —

- ☒ Spatial Attention Module — নাম আছে
- ☒ Multi-scale Fusion — নাম আছে
- ☒ Histogram equalization, color conversion — নাম আছে
- ☒ Augmentation — উল্লেখ আছে
- ☒ Grad-CAM, LIME — নাম আছে

তুমি এগুলোর নাম দেখেই বুঝেছ কী করে — তাই diagram এ সেগুলো box হিসেবে দেখিয়েছ। ভেতরে কীভাবে কাজ করে সেটা diagram এ দরকার নেই।

মূল কথা — ৩ লাইনে

১. Abstract এ যা explicitly লেখা আছে — সেটা diagram এ রাখো।
 ২. Abstract এ যে method এর নাম আছে — সেই method এর standard steps নিজে থেকে add করো।
 ৩. যা abstract এ নেই এবং তুমি জানো না — সেটা diagram এ আনার দরকার নেই।
-

1. Machine Learning Models

Category	Model Name	Key Focus / Use Case
Classification (Discrete Classes)	• Logistic Regression	Predicts categories or labels
	• Support Vector Machine (SVM)	(e.g., Spam vs Not Spam)
	• Decision Tree & Random Forest	
	• K-Nearest Neighbors (KNN)	
	• Naive Bayes	
Regression (Continuous Values)	• Linear Regression	Predicts continuous numbers
	• Polynomial Regression	(e.g., Price, Temperature)
	• Ridge & Lasso Regression	
	• Support Vector Regression (SVR)	

2. Deep Learning Models

Architecture Type	Key Models	Primary Application Area
ANN (Artificial Neural Network)	• Multilayer Perceptron (MLP)	Foundational architecture, excellent for Tabular Data
	• Deep Belief Networks (DBN)	
CNN (Convolutional Neural Network)	• LeNet-5	Computer Vision (Images, Videos, Pixel Patterns)
	• AlexNet	
	• VGG-16 / VGG-19	
	• ResNet	
	• Inception / GoogLeNet	

RNN (Recurrent Neural Network)	• Vanilla RNN • LSTM (Long Short-Term Memory) • GRU (Gated Recurrent Unit)	Sequential Data (Time-series, Stock Prices, Speech)
NLP (Natural Language Processing)	• Transformer • BERT (Google) • GPT (OpenAI) • T5 (Google)	Human Language (Text Translation, Chatbots, Generation)

Universal Literature Review Builder Pattern

মূল কথা আগে বোঝো

Literature review মানে একটা "গল্প" বলা:

আগে এই হয়েছিল → তারপর এটা হয়েছিল →

তারপর এটা → কিন্তু সবার সমস্যা ছিল →

তাই আমার paper এটা করল

এটাকে বলে **"Funnel Structure"** — broad থেকে narrow।

সবচেয়ে পুরনো / basic কাজ

↓

একটু advanced কাজ

↓

আরো advanced কাজ



সবচেয়ে recent / কাছের কাজ



সবার limitations একসাথে



তোমার paper এর contribution

STEP 1 — Abstract থেকে যা বের করতে হবে

যেকোনো abstract পেলে প্রথমে এই ৪টা জিনিস খুঁজে বের করো:

① TOPIC → কী বিষয়ে কাজ?

(Eye disease? Skin cancer? WBC? Crop disease?)

② METHOD → কোন technology propose করা হয়েছে?

(CNN? Transformer? IoT? Ensemble?)

③ KEY MODULES → কোন special feature আছে?

(Attention? Augmentation? XAI? Fusion?)

④ METRIC → কী দিয়ে evaluate করা হয়েছে?

(Accuracy? Recall? F1? AUC?)

এই ৪টা পেলে তুমি জানবে — কোন ধরনের **previous work cite** করতে হবে।

◆ STEP 2 — ৭টা Model/Paper কোন Pattern এ সাজাবে

এই ৭-step progression follow করো — সবসময় কাজ করে:

Paper 1 → Traditional / Classical method

(সবচেয়ে basic, পুরনো approach)

Paper 2 → Basic Deep Learning

(Simple CNN বা basic ML model)

Paper 3 → Transfer Learning / Pretrained Model

(VGG, ResNet, InceptionNet ইত্যাদি)

Paper 4 → Attention / Advanced Architecture

(Attention mechanism, LSTM, etc.)

Paper 5 → Transformer-based method

(ViT, Swin, MaxViT ইত্যাদি)

Paper 6 → Explainability / Interpretability

(XAI, Grad-CAM, LIME ইত্যাদি)

Paper 7 → Most Recent / Closest to your paper

(তোমার paper এর সবচেয়ে কাছের কাজ)

শেষে → **Gap Statement + তোমার paper**

◆ STEP 3 — প্রতিটা Paper এর Sentence Pattern

প্রতিটা paper এর জন্য এই **4-sentence formula**:

Sentence 1 → [Author] [verb] [method] for [problem/goal].

Sentence 2 → [Dataset / experiment details].

Sentence 3 → [Result — accuracy বা performance].

Sentence 4 → However, [limitation].

Verb list — এগুলো rotate করে use করো:

proposed / developed / introduced / presented / utilized / applied / employed /
investigated / designed / implemented

Limitation Phrases — এগুলো rotate করো:

However, the model... Despite these results, the approach... Nevertheless, the
system... Although effective, this method...

◆ STEP 4 — Limitation এর ভাণ্ডার

Abstract এর topic অনুযায়ী এই limitations use করো:

Accuracy related:

- ...achieved limited accuracy and failed to generalize across diverse datasets.
- ...showed lower performance on multi-class classification tasks.

Data related:

- ...was trained on a small and homogeneous dataset, limiting its generalizability.
- ...relied on a single publicly available dataset, reducing clinical applicability.

Architecture related:

- ...used a standard architecture without domain-specific optimization.
- ...lacked attention mechanisms to focus on task-relevant regions.
- ...could not capture long-range spatial dependencies within images.

Interpretability related:

- ...did not incorporate any explainability mechanism, limiting clinical trust.
- ...provided no visual explanation for model decisions.

Scope related:

- ...was limited to binary classification and could not handle multiple categories.
- ...was not validated in real-world or cross-institutional settings.

Computational related:

- ...required high computational resources, limiting practical deployment.
 - ...was not optimized for real-time or mobile applications.
-

STEP 5 — Gap Statement Formula

Literature review এর শেষে সবসময় এই structure:

"Despite [what has been done], existing approaches
suffer from [common limitations].

[Specific problem 1], [specific problem 2],
and [specific problem 3] remain unresolved.

The proposed [your method] addresses these gaps by
[what your paper does], achieving [your result]."

 এখন দেখো — **Abstract** দিলে কীভাবে **Apply**
করবে

Example: Eye Disease (EDDNet30 Abstract)

Step 1 থেকে বের করলাম:

- Topic → Eye disease / Fundus image classification
- Method → Custom CNN (EDDNet30)
- Key Modules → Spatial Attention, Multi-scale Fusion, XAI
- Metric → Accuracy (95.29%)

Step 2 অনুযায়ী ৭টা paper এর type:

Paper 1 → Traditional image processing for eye disease

Paper 2 → Basic CNN for fundus image

Paper 3 → Transfer learning (VGG/ResNet) for eye disease

Paper 4 → Attention-based model for retinal detection

Paper 5 → Multi-scale feature extraction

Paper 6 → XAI in medical imaging

Paper 7 → Most recent multi-class eye disease classifier

Step 3 এর formula apply করে লেখা:

[Paper 1 — Traditional] Sinthanayothin et al. proposed a traditional image processing approach for retinal lesion detection using morphological operations and thresholding techniques. The method was tested on a small set of fundus photographs. While the approach demonstrated early feasibility of automated retinal analysis, it relied heavily on handcrafted rules and failed to generalize across diverse image conditions and disease types.

[Paper 2 — Basic CNN] Gulshan et al. developed a basic deep CNN for diabetic retinopathy detection trained on a large fundus image dataset. The model achieved high sensitivity for binary classification of referable and non-referable cases. However, the system was designed exclusively for diabetic retinopathy and could not classify multiple retinal disease categories simultaneously.

[Paper 3 — Transfer Learning] Li et al. applied transfer learning using pretrained VGG16 and ResNet50 architectures for retinal disease classification. Although transfer learning reduced training time and performed reasonably well on benchmark datasets, these standard architectures lacked mechanisms to focus on spatially relevant retinal regions, resulting in lower performance on subtle and complex disease patterns.

[Paper 4 — Attention] Orlando et al. introduced an attention-based deep learning model for glaucoma detection from fundus images. The attention

mechanism allowed the model to concentrate on diagnostically relevant retinal structures. Nevertheless, the model was limited to single-disease detection and did not integrate multi-scale feature extraction, reducing its applicability for multi-class classification.

[Paper 5 — Multi-scale] Jiang et al. employed a multi-scale convolutional framework for retinal lesion segmentation, capturing features at different spatial resolutions. The approach improved detection of both fine-grained and coarse lesion structures. Despite this, the model did not incorporate spatial attention modules, limiting its ability to suppress irrelevant background features in complex fundus images.

[Paper 6 — XAI] Chetoui and Akhloufi applied explainable AI techniques including Grad-CAM to interpret CNN-based eye disease classifiers, providing visual explanations for model predictions. While interpretability was improved, the underlying classification models were standard architectures without specialized attention or multi-scale modules, constraining diagnostic accuracy.

[Paper 7 — Most Recent] Zhao et al. proposed a deep learning framework for multi-class retinal disease classification using an ensemble of CNN models trained on diverse fundus image datasets. The model achieved competitive accuracy across several disease categories. However, the ensemble approach increased computational complexity and lacked integrated explainability mechanisms necessary for transparent clinical deployment.

[Gap Statement] Despite these advancements, existing approaches suffer from key limitations including inability to perform robust multi-class classification, insufficient spatial focus on disease-relevant retinal regions, limited multi-scale feature integration, and absence of built-in explainability. The proposed EDDNet30 model addresses these gaps through its novel 30-layer architecture incorporating spatial attention and multi-scale fusion modules, achieving 95.29% accuracy across 9 disease categories with integrated XAI-based interpretability.

Quick Cheat Sheet — Exam এ হাতে রাখো

STEP 1: Abstract পড়ে → Topic, Method, Module, Metric বের করো

STEP 2: ৭টা paper এর type ঠিক করো

Traditional → Basic DL → Transfer Learning →

Attention → Transformer → XAI → Most Recent

STEP 3: প্রতিটায় 4-sentence formula

[Author] + [Method] + [Result] + "However, [Limitation]"

STEP 4: Limitation ঘুরিয়ে ফিরিয়ে use করো

(accuracy / data / architecture / interpretability / scope)

STEP 5: Gap Statement দিয়ে শেষ করো

"Despite... existing methods fail to..."

The proposed study addresses..."
